General Mechanics

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Abstract

This work presents a framework for mechanics derived from a single principle: the generalized Stokes identity on an informational manifold. By synthesizing this geometric bookkeeping with principles from thermodynamics and information theory, the framework demonstrates how core structures of physics can be reconstructed. The analysis yields the canonical rate equations of mechanics, the inverse–area scaling of boundary fields, and the propagation laws for wave-like transport. The same structure yields operational limits such as time–information complementarity and finite channel capacity. For operational use, a constant-free, one-datum calibration maps the underlying informational field to measured sectors (e.g., gravitational/thermal, electric), rendering the framework predictive while keeping its core purely manifold-theoretic.

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1 Introduction

1.1 Aim and stance

This work takes *General Mechanics* to mean: familiar mechanics appear as *relational motifs* read from a single geometric identity (the generalized Stokes identity) on a chosen *informational* manifold. Relations are primary; observables are relational motifs. An observer specifies a region $U \subset M$ with boundary ∂U; motifs are read from the boundary.

1.2 Units and calibration

Kz units App. A are used so energies and powers are rates. The abstract informational field is mapped to measured sectors (gravitational, electric, thermal) by a *one-datum* linear calibration; no bare constants are carried through the algebra.

1.3 How to use the framework (recipe)

1. Declare $U \subset M$ and ∂U (boundary or sector).
2. Write the Stokes identity $d(*j) = \sigma\,\mathrm{vol}$ and, when needed, the *cut identity* on U (2.5), (2.4).
3. Choose a closure suited to the regime: **frame** (2.9) for frame problems 4.1.1, 4.1.2; or a **wave-supporting** closure for export on source-free segments §3. If the window is LTI and causal, the Causality–Dispersion comparison may be used locally §3.4.
4. Count distinctions on ∂U in a ternary alphabet (*yes/no/unknown*) using information-theoretic bookkeeping, and relate them to j.
5. If comparison with a measured sector is required, fix a single datum to calibrate App. A.
6. Read **frame laws** from the inverse-area rule ($\propto 1/A$; 4.1.1, 4.1.2; for wave export and far-field envelopes §3.
7. Apply operational limits: time–information complementarity 4.2.1 and the counting-noise floor 4.2.2 for finite windows/areas.

1.4 Scope and reductions

Dimensional reductions (1D/2D confinement) are handled by replacing $A_n(r)$ with an effective cross-section/circumference; the resulting **frame** and wave summaries appear in §4.6.2. Detailed wave envelopes and exported-power statements are collected in §3.

Literature note. Differential forms/Stokes background: Frankel [1]. Poynting/export in EM: Jackson [2], Landau–Lifshitz–Pitaevskii [3]. Information/capacity: Shannon [4], Gallager [5]. Erasure cost: Landauer [6], Bennett [7].

2 Axioms

Three pillars (thermodynamics, information–relational, Stokes).

(i) ***Thermodynamics as content.*** The piston thought–experiment generalizes via Stokes.

(ii) ***Information theory / relational.*** Counting and priors are Shannon-based over a *ternary* alphabet (*yes/no/unknown*); observables are relational.

(iii) ***Geometric calculus (Stokes identity).*** A single generalized Stokes identity (2.1) is applied on a chosen region and its time extension, with cap/side (*worldtube*) decomposition.

2.1 Core Axioms

2.1.1 Relational

Relational primacy. Relations are the primitives; observables are relational. A model specifies a region $U \subset M$ and its boundary ∂U.

Counting & priors (trits/nats). A configuration determined by n ternary distinctions has $W = 3^n$ microstates. With a prior $p(x)$ on an alphabet $\mathcal{A}$ of size 3, the uncertainty is the Shannon entropy $H(p) = -\sum_{x\in\mathcal{A}} p(x) \ln p(x)$ (nats), with

$$H_{\text{trits}} = \frac{H_{\text{nats}}}{\ln 3}.$$

2.1.2 Identity (Stokes postulate)

The informational/entropic flux 1-form j (with Hodge-dual $j^{(n)}$) and production $\Sigma := \sigma \,\text{vol}$ satisfy the postulate

$$d\, j^{(n)} = \Sigma \quad \Longleftrightarrow \quad \partial_\mu J^\mu = \sigma. \tag{2.1}$$

Boundary identification and heat mapping. On boundaries, identify the information current with the entropy current $j \equiv j_S$ (counts/area/time), and the volume density $i \equiv s$ (nats/vol). With a boundary temperature field T_{z}, the heat flux is $q := T_{\text{z}}\, j_S$:

$$j \equiv j_S, \qquad i \equiv s, \qquad q := T_{\text{z}}\, j_S. \tag{2.2}$$

2.2 Key Consequences of the Axioms

2.2.1 Continuity forms (derived)

From (2.1) on $U \times [t_0, t_1]$,

$$\int_{\partial(U\times[t_0,t_1])} j^{(n)} \;=\; \int_{U\times[t_0,t_1]} \Sigma. \tag{2.3}$$

On a fixed U, the cut identity is

$$\dot{I}_{\text{enc}}(t) = \Phi_{\text{in}}(t) - \Phi_{\text{out}}(t) + \Pi(t), \tag{2.4}$$

where I_{enc}, $\Phi_{\text{in/out}}$, and Π denote the volume/boundary integrals of i, $(\mathbf{j}\cdot\mathbf{n})_{\mp}$, and σ. In local coordinates,

$$\partial_t i(x,t) + \nabla\cdot \mathbf{j}(x,t) = \sigma(x,t). \tag{2.5}$$

2.2.2 Relational Equation of State (REOS)

For admissible boundary/control variations $R^{(a)}$ and calibrated rate density $\rho = \rho(R)$, the price one-form $\lambda := \lambda_a\, dR^{(a)}$ (with $\lambda_a := -\,\partial\rho/\partial R^{(a)}$) satisfies

$$d\rho \;=\; -\,\lambda_a\, dR^{(a)}. \tag{2.6}$$

If $d\lambda = 0$, ρ is path-independent. If $d\lambda \neq 0$, cyclic variations entail nonnegative production σ in (2.1)–(2.4).

2.2.3 Canonical rates from Stokes on (q, t)

For a rate (Poincaré–Cartan) 1-form on configuration–time space

$$\Theta_{\mathcal{R}} = P_i\, dq^i - H(q, P, t)\, dt, \tag{2.7}$$

evaluating Stokes on a rectangular 2-surface Σ spanned by two nearby paths gives

$$\int_{\partial\Sigma} \Theta_{\mathcal{R}} \;=\; \int_{\Sigma} d\Theta_{\mathcal{R}} \;=\; 0,$$

which is equivalent to

$$\dot{q}^i = \partial_{P_i} H, \qquad \dot{P}_i = -\,\partial_{q^i} H. \tag{2.8}$$

2.3 Constitutive Models

Linear channel closure. In linear, isotropic, near-equilibrium regimes,

$$\mathbf{E}_I := -\,\nabla\Phi_I, \qquad \mathbf{j} = -\,\kappa\,\mathbf{E}_I. \tag{2.9}$$

Wave-supporting closure. On a window $[t_0, t_1]$, fields (U, V) with a linear, causal constitutive map

$$V = \chi : U, \tag{2.10}$$

have stored energy

$$u \;:=\; \tfrac{1}{2}\, U{:}V, \tag{2.11}$$

and a spatial flux $\mathbf{S}$ satisfying

$$\partial_t u \;+\; \nabla\cdot \mathbf{S} \;=\; -\,\Pi, \qquad \Pi \geq 0. \tag{2.12}$$

3 Universal wave transport & refraction

3.1 Wave-supporting closure and exported power

Integrating the local balance (2.12) on a source-free, lossless subregion W and applying Stokes on its spatial boundary $\Sigma(r)$ yields radius-independent export

$$P(r) := \int_{\Sigma(r)} \langle \mathbf{S} \rangle \cdot \mathbf{n}\, \mathrm{d}A \;=\; \text{const. in } r, \tag{3.1}$$

and therefore the far-field envelope scales as

$$\big\langle |U(r)| \big\rangle \;\propto\; [A(r)]^{-1/2}, \tag{3.2}$$

where $A(r)$ is the area of the boundary $\Sigma(r)$. With attenuation μ along a *worldtube*, $P(r) \propto \exp\big[-\int \mu\, \mathrm{d}s\big]$ and the envelope acquires

$$\big\langle |U(r)| \big\rangle \;\propto\; [A(r)]^{-1/2} \exp\left[-\tfrac{1}{2}\int \mu\, \mathrm{d}s\right]. \tag{3.3}$$

For confined propagation, replace $A(r)$ by the effective cross-section $A_\perp$ of the transport tube, giving

$$\big\langle |U| \big\rangle \;\propto\; [A_\perp]^{-1/2}. \tag{3.4}$$

3.2 Eikonal limit, Fermat principle, and tube transport

Assume a high-frequency limit on a homogeneous or slowly varying window: the amplitude and index vary slowly relative to the carrier,

$$\|\nabla A\|/|A| \ll \omega, \qquad \|\nabla n\|/n \ll \omega/c_I.$$

Use the ansatz

$$U(x,t) \;\sim\; \mathrm{Re}\Big\{A(x)\, e^{i\omega S(x) - i\omega t}\Big\}, \tag{3.5}$$

with slowly varying amplitude A and phase S (eikonal). The constitutive description (2.10) defines an *effective index* $n(x)$ from dispersion and a local speed $c_{\text{eff}} := c_I/n(x)$. In locally isotropic media (or for an effectively isotropic eigenmode), n is a scalar; otherwise n denotes the scalar slowness along the ray. At leading order, S satisfies the Hamilton–Jacobi equation

$$|\nabla S(x)|^2 \;=\; n^2(x)\,\frac{\omega^2}{c_I^2}, \tag{3.6}$$

and rays are the integral curves of $\mathbf{k} := \nabla S$ with unit tangent $\hat{\mathbf{s}} := \mathbf{k}/|\mathbf{k}|$; they extremize the optical time

$$\delta \int_\gamma \frac{n(x)}{c_I}\,\mathrm{d}s \;=\; 0, \tag{3.7}$$

yielding Snell's law at interfaces. The ray equation is, equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}s}\big(n\,\hat{\mathbf{s}}\big) \;=\; \nabla n, \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}\hat{\mathbf{s}}}{\mathrm{d}s} \;=\; \nabla_\perp\big(\ln n\big). \tag{3.8}$$

Tube transport for the slowly varying amplitude gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\Big(|A|^2 A_\perp\Big) \;=\; -\,|A|^2 A_\perp\,\mu(s), \qquad \Rightarrow \qquad |A|^2 A_\perp \propto \exp\!\left[-\int \mu\,\mathrm{d}s\right], \tag{3.9}$$

so $|A| \propto A_\perp^{-1/2}$ (lossless), consistent with (3.2).

3.3 Nonlinear refraction from constitutive feedback

If the constitutive map χ depends on a local invariant of the field, e.g. the energy density $u = \frac{1}{2}\,U{:}V$ or occupancy on the window, then the effective index becomes *amplitude-dependent*:

$$n \;=\; n_0 \;+\; \alpha\,u \;+\; \mathcal{O}(u^2) \qquad \text{(weak feedback)}. \tag{3.10}$$

Using (3.9), $u \propto |A|^2 \propto A_\perp^{-1}$ (lossless), so the ray equation acquires a self-consistent bending term:

$$\frac{\mathrm{d}}{\mathrm{d}s}\big(n\,\hat{\mathbf{s}}\big) \;=\; \nabla n\big(u(x)\big) \quad \Longleftrightarrow \quad \frac{\mathrm{d}\hat{\mathbf{s}}}{\mathrm{d}s} \;=\; \nabla_\perp\big(\ln n(u)\big), \tag{3.11}$$

closing the loop geometry $\to |A| \to u \to n(u) \to$ geometry. In differential-geometric language, the expansion $\theta := \mathrm{d}(\ln A_\perp)/\mathrm{d}s$ along a congruence of rays follows a Raychaudhuri-type identity

$$\frac{\mathrm{d}\theta}{\mathrm{d}s} \;=\; -\,\tfrac{1}{2}\,\theta^2 \;-\; \varsigma^2 \;-\; \mathcal{R}_{\text{eff}}[n(u)] \;+\; \text{attenuation terms}, \tag{3.12}$$

where ς denotes the shear magnitude and $\mathcal{R}_{\text{eff}}[n(u)]$ collects the transverse curvature induced by gradients of $\ln n(u)$ along the congruence (vanishes for uniform n). Equation (3.12) is the refractive focusing identity induced by the feedback $n{\circ}u{\circ}A_\perp$.

3.4 Causality–Dispersion (KK-window corollary)

Hypotheses (declared on this window). (H1) Linear, time-invariant response on the window: $V = \chi * U$. (H2) Causality: $\chi(t) = 0$ for $t < 0$ with $\chi \in L^1(\mathbb{R})$. (H3) Regularity: $\widehat{\chi}(\omega)$ exists and is bounded. (These are the window-level specializations corresponding to ⋆2.3.)

Witness (mathematics only). Under (H1)–(H3), the boundary values of $\widehat{\chi}$ on the real axis satisfy the Hilbert-pair relations

$$\operatorname{Re}\widehat{\chi}(\omega) = \mathcal{H}\{\operatorname{Im}\widehat{\chi}\}(\omega), \qquad \operatorname{Im}\widehat{\chi}(\omega) = -\mathcal{H}\{\operatorname{Re}\widehat{\chi}\}(\omega),$$

and, for minimum-phase windows,

$$\arg\widehat{\chi}(\omega) = -\mathcal{H}\{\ln|\widehat{\chi}|\}(\omega).$$

Derivation (sketch). Causality and L^1 imply analyticity of $\widehat{\chi}$ in the upper half-plane; the Plemelj–Sokhotski formulas yield the Hilbert-pair relations. Only the declared window choices and complex analysis are used.

Falsification (same window). On this window, measure Re / Im (or $|\cdot|$ / arg) of $\widehat{\chi}$; a persistent Hilbert-pair residual beyond tolerance falsifies (H1)–(H3) (or the minimum-phase clause) on this window.

3.5 Worldtubes, moving boundaries, and boundary sectors

For moving boundaries, the side integrand is $\mathbf{j}\cdot\mathbf{n} - i\,v_n$ App. B. For fixed boundary sectors, export remains constant; envelopes follow $[A_\perp]^{-1/2}$ with $A_\perp$ the sector's cross-section.

4 Corollaries

Consequences of ⋆2.2.1 (and, where invoked, (2.5)). Wave-window items cite the wave closure and balance ⋆2.3.

4.1 Continuity

4.1.1 Info–Gas & Stokes

Setup. A narrow tube $\mathcal{T}$ has side tangent to $\mathbf{j}$ and cross-sections $\Sigma(r)$ orthogonal to its axis. Work on a frame with $\partial_t i = 0$; the tube lies in an open $W \subset U$ with $\sigma|_W = 0$.

Derivation. From the local continuity identity (2.5) on W, the directed flux through $\Sigma(r)$ is independent of r:

$$\Phi(r) := \int_{\Sigma(r)} \mathbf{j}\cdot\mathbf{n}\,\mathrm{d}A \;=\; \Phi \;\text{(const.)}. \tag{4.1}$$

Hence the cross-sectional mean of the normal component satisfies

$$\langle j_n\rangle(r) \;=\; \frac{\Phi}{A(r)}, \qquad A(r) := \int_{\Sigma(r)} \mathrm{d}A. \tag{4.2}$$

Specialization. For a conical sector of fixed solid angle $\Delta\Omega$ in n spatial dimensions,

$$A_n(r) = \Delta\Omega\, r^{\,n-1} \;\Rightarrow\; \langle j_n\rangle(r) = \frac{\Phi}{\Delta\Omega}\, r^{-(n-1)}. \tag{4.3}$$

Directional filtering (inward component). With $\Phi_- := \int_{\Sigma(r)} (\mathbf{j}\cdot\mathbf{n})_-\, \mathrm{d}A$, relations (4.2)–(4.4) hold under the substitution $\Phi \mapsto \Phi_-$.

Calibration. With the Linear Channel Closure (2.9),

$$\big|\mathbf{E}_I(r)\big| = \frac{1}{\kappa}\,\big\langle j_n\big\rangle(r) = \frac{1}{\kappa}\,\frac{\Phi}{A(r)}. \tag{4.4}$$

Falsification. On a source-free tube, if $\big\langle j_n\big\rangle(r)\,A(r)$ deviates from a constant by more than a stated tolerance over a declared r-range, the premises of this corollary are contradicted.

Provenance. (2.5), (2.9), (4.2); directional split App. B.

Literature note. Flux conservation from divergence-free fields: Frankel [1]. Inverse-power radial profiles and solid-angle sectors: Jackson [2].

4.1.2 Frame boundary (inverse-power fields)

Setup. Work on a frame with $\partial_t i = 0$ on an open annulus $W \subset U$ where $\sigma|_W = 0$. Let $\Sigma(r) \subset W$ be nested spherical boundaries of radius r with area $A_n(r)$. Define the directed flux $\Phi(r) := \int_{\Sigma(r)} \mathbf{j}\cdot\mathbf{n}\,\mathrm{d}A$.

Derivation. From the Stokes identity in integral/cut form (2.3)-(2.4) on W, $\Phi(r)$ is independent of r. Therefore

$$\big\langle j_n\big\rangle(r) = \frac{\Phi}{A_n(r)}. \tag{4.5}$$

With the Linear Channel Closure (2.9), the frame amplitude satisfies

$$\big|\mathbf{E}_I(r)\big| = \frac{1}{\kappa}\,\frac{\Phi}{A_n(r)}. \tag{4.6}$$

Directional filtering (inward component). With $\Phi_- := \int_{\Sigma(r)} (\mathbf{j}\cdot\mathbf{n})_-\, \mathrm{d}A$ (directional split App. B), relations (4.5)–(4.6) hold under the substitution $\Phi \mapsto \Phi_-$.

Specialization. In $n = 3$, $A_3(r) = 4\pi r^2$, giving

$$\big|\mathbf{E}_I(r)\big| = \frac{\Phi}{4\pi\,\kappa}\,\frac{1}{r^2}. \tag{4.7}$$

Calibration. With linear sector mappings $g = c_M\,|\mathbf{E}_I|$ and $E = c_Q\,|\mathbf{E}_I|$,

$$g(r) = \frac{c_M\,\Phi}{4\pi\,\kappa}\,\frac{1}{r^2}, \qquad E(r) = \frac{c_Q\,\Phi}{4\pi\,\kappa}\,\frac{1}{r^2}. \tag{4.8}$$

A single datum $(r_\star, Y_\star)$ in a given sector yields the constant–free ratio

$$Y(r) = Y_\star\left(\frac{\Phi}{\Phi_\star}\right)\left(\frac{r_\star}{r}\right)^2, \qquad Y \in \{g, E\}. \tag{4.9}$$

Falsification. On a source-free frame, if $r^2\big|\mathbf{E}_I(r)\big|$ departs from a constant by more than a declared tolerance over a stated r-range, the premises of this corollary are contradicted.

Provenance. (2.3), (2.4), (2.9), (4.5), App. B.

Literature note. Frankel [1]; Jackson [2].

4.2 Time–Information

4.2.1 Time–information complementarity

Setup. A boundary patch $A \subset \partial U$ is observed over a window $[t, t+\Delta t]$. With the directional split, the inward, window-averaged counting rate is

$$J_A \;:=\; \frac{1}{\Delta t}\int_t^{t+\Delta t}\int_A \big(\mathbf{j}\cdot\mathbf{n}\big)_-\,\mathrm{d}A\,\mathrm{d}t \quad \text{(nats/time)}. \tag{4.10}$$

An erasure power budget P acts at boundary temperature T_{z}. Reliable channel capacities are denoted $C_{\mathrm{chan}}^{(\mathrm{nats})}(\varepsilon)$ and $C_{\mathrm{chan}}^{(\mathrm{trits})}(\varepsilon)$.

Derivation. By boundary identification and heat mapping (on boundaries $j \equiv j_S$ and $q := T_{\mathrm{z}}\, j_S$), the erasure-limited rate bound is $P_{\mathrm{z}}/T_{\mathrm{z}}$ (nats/time). Combining this with the inward window rate (4.10) and any ε-reliable channel capacity gives:

$$\frac{\Delta I_{\mathrm{nats}}}{\Delta t} \;\le\; \min\Big\{\, J_A,\; C_{\mathrm{chan}}^{(\mathrm{nats})}(\varepsilon),\; \tfrac{P_{\mathrm{z}}}{T_{\mathrm{z}}} \,\Big\}. \tag{4.11}$$

Specialization. Using the base-3 to natural-log conversion 1 trit $= \ln 3$ nats,

$$\frac{\Delta I_{\mathrm{trits}}}{\Delta t} \;\le\; \min\Big\{\, \tfrac{J_A}{\ln 3},\; C_{\mathrm{chan}}^{(\mathrm{trits})}(\varepsilon),\; \tfrac{P_{\mathrm{z}}}{T_{\mathrm{z}}\ln 3} \,\Big\}. \tag{4.12}$$

Calibration. For one reliable trit ($\Delta I_{\mathrm{trits}} = 1$), the minimal window duration obeys

$$\Delta t \;\ge\; \max\Big\{\, \big(J_A/\ln 3\big)^{-1},\; C_{\mathrm{chan}}^{(\mathrm{trits})}(\varepsilon)^{-1},\; (\ln 3)\,T_{\mathrm{z}}/P_{\mathrm{z}} \,\Big\}. \tag{4.13}$$

Falsification. On a fixed patch A with measured $(J_A, P_{\mathrm{z}}, T_{\mathrm{z}})$ and capacity target ε, if a protocol achieves a one-trit decision with Δt smaller than the bound in (4.13) by $> 5\%$ under the same budgets, then at least one premise (capacity estimate, boundary mapping, or rate accounting) is violated on that window.

Provenance. ⋆2.1.2, (4.10), (4.11), (4.12), (4.13), App. B.

Literature note. Shannon [4]; Gallager [5]; Landauer [6]; Bennett [7].

4.2.2 Noise on boundary windows

Setup. A boundary patch $A \subset \partial U$ is observed over a window $[t, t+\Delta t]$. Window counts are $N_A(\Delta t)$ with windowed Fano factor $F_A := \mathrm{Var}(N_A)/\mathbb{E}[N_A]$. Define the inward patch integral and patch-mean density

$$J_A := \int_A \big(\mathbf{j}\cdot\mathbf{n}\big)_-\,\mathrm{d}A, \qquad j_A := \frac{\mathbb{E}[N_A]}{|A|\,\Delta t} = \frac{J_A}{|A|}.$$

An effective duration $\Delta t_{\mathrm{eff}} \le \Delta t$ is obtained from the count autocorrelation on the same record.

Derivation. From the Linear Channel Closure (2.9) ($\mathbf{j} = -\kappa\,\mathbf{E}_I$), the variance floor of any unbiased estimator $\widehat{E}_I$ over the window satisfies

$$\mathrm{Var}(\widehat{E}_I) \;\gtrsim\; \frac{F_A\, j_A}{\kappa^2\,|A|\,\Delta t_{\mathrm{eff}}}. \tag{4.14}$$

Specialization (correlated arrivals). If arrivals have a correlation time $\tau > 0$, the effective duration scales as $\Delta t_{\text{eff}} \sim \Delta t/\tau$, giving

$$\operatorname{Var}(\widehat{E}_I) \;\gtrsim\; \frac{j_A}{\kappa^2\,|A|}\,\frac{\tau}{\Delta t}. \tag{4.15}$$

Calibration. With a one–datum sector mapping $Y = c_{\text{sec}}\,|E_I|$ on the same region, the measured-unit floor is

$$\operatorname{Var}(\widehat{Y}) \;\gtrsim\; \frac{c_{\text{sec}}^2\,F_A\,j_A}{\kappa^2\,|A|\,\Delta t_{\text{eff}}}. \tag{4.16}$$

In a far-field frame, the inverse-area law (4.2) and a source family labeled by total energy E_{src} with $\Phi \propto E_{\text{src}}$ imply the fractional mapping

$$\frac{\delta Y(r,t)}{Y(r)} = \frac{\delta\Phi}{\Phi} = \frac{\delta E_{\text{src}}(t)}{E_{\text{src}}}. \tag{4.17}$$

Hence the one-sided PSD and in-band rms specializations are

$$S_g(\omega) = \left(\frac{g(r)}{E_{\text{src}}}\right)^2 S_E(\omega), \tag{4.18}$$

$$g_{\text{rms}}(r) \;\approx\; g(r)\,\frac{\sqrt{\displaystyle\int_\Omega S_E(\omega)\,\mathrm{d}\omega}}{E_{\text{src}}} \;=\; g(r)\,\frac{\sqrt{\operatorname{Var}[E]_\Omega}}{E_{\text{src}}}. \tag{4.19}$$

Falsification. In a shot-noise regime $F_A \to 1$ with $\Delta t_{\text{eff}} \to \Delta t$, hold j_A and $|A|$ fixed and vary Δt over at least one decade; if

$$\operatorname{Var}(\widehat{Y})\,\Delta t$$

deviates from the constant predicted by (4.16) by more than a factor of 3, the closure or counting assumptions used to obtain (4.14) are falsified on that window.

Provenance. (2.9), (4.14), (4.15), (4.16), (4.2), (4.17), (4.18), (4.19).

Literature note. Hu–Verdaguer (stochastic gravity) and Kingman (Poisson processes).

4.3 Thermodynamics

4.3.1 Continuity & Clausius

Setup. Let $V \subset U$ be a fixed region with outward unit normal $\mathbf{n}$. On a boundary patch $A \subset \partial V$ over a window, denote the boundary entropy flux density by $\mathbf{j}_S$.

Boundary heat. On $[t_0, t_1]$,

$$Q_A([t_0,t_1]) \;=\; \int_{t_0}^{t_1}\!\!\int_A T_{\text{z}}\,\mathbf{j}_S\cdot\mathbf{n}\,\mathrm{d}A\,\mathrm{d}t, \tag{4.20}$$

and the frame heat-flux density is

$$\mathbf{q} \;=\; T_{\text{z}}\,\mathbf{j}_S. \tag{4.21}$$

Derivation. From the local continuity identity (2.5) and the boundary identification/heat mapping (by (2.1.2), $j \equiv j_S$, $q := T_{\text{z}} S$), the Clausius local form follows.

Clausius form (local). With $s = i$ and $\mathbf{j}_S = \mathbf{q}/T_{\mathrm{z}}$,

$$\partial_t s \;+\; \nabla\cdot\left(\frac{\mathbf{q}}{T_{\mathrm{z}}}\right) \;=\; \sigma \;\geq\; 0. \tag{4.22}$$

Clausius form (cut). From the cut identity (2.4), using the directional split,

$$\dot{S}_{\mathrm{enc}} \;=\; \int_{\partial V} \frac{(\mathbf{q}\cdot\mathbf{n})_-}{T_{\mathrm{z}}}\,\mathrm{d}A \;-\; \int_{\partial V} \frac{(\mathbf{q}\cdot\mathbf{n})_+}{T_{\mathrm{z}}}\,\mathrm{d}A \;+\; \Pi_{th} \;\geq\; 0, \tag{4.23}$$

with $\Pi_{th} := \int_V \sigma\,\mathrm{d}^n x$ (by (2.1.2), $j \equiv j_S$, $q := T_{\mathrm{z}} S$).

Calibration to the informational field. With the Linear Channel Closure (2.9) and the one–datum sector calibration,

$$\mathbf{j} \;=\; -\,\kappa\,\mathbf{E}_I, \qquad \mathbf{q} \;=\; -\,T_{\mathrm{z}}\,\kappa\,\mathbf{E}_I. \tag{4.24}$$

Falsification. On a closed boundary with measured $\mathbf{q}, T_{\mathrm{z}}$ and access to $\dot{S}_{\mathrm{enc}}$, any reproducible violation of (4.22) or a negative right–hand side in (4.23) beyond experimental uncertainty falsifies the premises (local identity (2.5) and/or the boundary mapping used).

Provenance. (2.5), (2.4), (2.1.2), (2.9), (4.22), (4.23).

Literature note. Continuity in differential-form language and Stokes' theorem: Frankel [1]. Clausius inequality in macroscopic thermodynamics: Callen [8]; nonequilibrium balance form: de Groot & Mazur [9].

4.3.2 External drives, delay & production

Setup. A narrow tube $\mathcal{T}$ connects an inlet cross–section Σ_{in} to an outlet Σ_{out}. The directed inlet/outlet fluxes (directional split) are

$$\Phi_{\mathrm{in}}(t) := \int_{\Sigma_{\mathrm{in}}} (\mathbf{j}\cdot\,\mathbf{n})_-\,\mathrm{d}A, \qquad \Phi_{\mathrm{out}}(t) := \int_{\Sigma_{\mathrm{out}}} (\mathbf{j}\cdot\,\mathbf{n})_+\,\mathrm{d}A.$$

Let $I_{\mathcal{T}}(t) := \int_{\mathcal{T}} i\,\mathrm{d}^n x$, $\Pi_{\mathcal{T}}(t) := \int_{\mathcal{T}} \sigma\,\mathrm{d}^n x$, and $A(s)$ the cross–section area at coordinate s.

Derivation. From the cut identity (2.4) applied to $\mathcal{T}$,

$$\dot{I}_{\mathcal{T}}(t) \;=\; \Phi_{\mathrm{in}}(t) - \Phi_{\mathrm{out}}(t) \;+\; \Pi_{\mathcal{T}}(t). \tag{4.25}$$

Transit and delay (transparent segment). On a source–free open subsegment $W \subset \mathcal{T}$ with $\sigma|_W = 0$ and directed speed profile $v(s) > 0$ along tube coordinate $s \in [0, L]$, the transit time is

$$\tau \;=\; \int_0^L \frac{\mathrm{d}s}{v(s)}. \tag{4.26}$$

With negligible dispersion,

$$\Phi_{\mathrm{out}}(t) \;=\; \Phi_{\mathrm{in}}(t-\tau). \tag{4.27}$$

For mild dispersion, $\Phi_{\mathrm{out}} = K * \Phi_{\mathrm{in}}$ with a causal kernel K of unit mass and mean τ.

Attenuation and production. In an advective regime with speed $v(s)$ and local removal rate $\mu(s) \geq 0$ (per unit length), the directed tube flux $\Phi(s,t) := \int_{\Sigma(s)} (\mathbf{j}\cdot\mathbf{n})_+ \, \mathrm{d}A$ satisfies

$$\partial_t \Phi(s,t) + v(s)\, \partial_s \Phi(s,t) \;=\; -\,\mu(s)\, \Phi(s,t). \tag{4.28}$$

Along characteristics,

$$S \;=\; \exp\Big(-\int_0^L \mu(s)\,\mathrm{d}s\Big), \qquad \tau \;=\; \int_0^L \frac{\mathrm{d}s}{v(s)}, \tag{4.29}$$

and

$$\Phi_{\mathrm{out}}(t) \;=\; S\, \Phi_{\mathrm{in}}(t-\tau). \tag{4.30}$$

Amplitude profile. With $\langle j_n\rangle(s,t) := \Phi(s,t)/A(s)$ the mean normal flux density, the transparent case ($S = 1$) gives $\langle j_n\rangle \propto 1/A(s)$. With attenuation,

$$\langle j_n\rangle(s,t) \;=\; \frac{\Phi(0, t-\tau_s)}{A(s)}\, \exp\Big(-\int_0^s \mu(u)\,\mathrm{d}u\Big), \qquad \tau_s = \int_0^s \frac{\mathrm{d}u}{v(u)}. \tag{4.31}$$

Series composition. For piecewise-constant segments $\{(L_k, v_k, \mu_k)\}$,

$$\tau \;=\; \sum_k \frac{L_k}{v_k}, \qquad S \;=\; \prod_k e^{-\mu_k L_k} \;=\; \exp\Big(-\sum_k \mu_k L_k\Big). \tag{4.32}$$

Kinematic invariance. If $\Sigma(s,t) = \Sigma(s - \xi(t))$ with $\dot\xi = U(t)$, the Stokes identity (2.3) on $\mathcal{T} \times [t_0, t_1]$ yields the same balance (4.25). A co-moving frame replaces ∂_t by the material derivative along $U(t)$ without altering (4.27)–(4.32).

Calibration. With the Linear Channel Closure (2.9) and the one–datum sector calibration, measured sector readouts Y satisfy

$$Y(s,t) = \frac{c_{\mathrm{sec}}}{\kappa}\, \langle j_n\rangle(s,t).$$

Falsification. With known $\{v(s), \mu(s)\}$ and measured inlet/outlet histories, any systematic failure of (4.27) (transparent) or (4.30) (attenuating) beyond experimental tolerance falsifies the advective transport premise (4.28) on the declared window.

Provenance. (2.4), (2.3), (2.9), (4.25)–(4.32).

Literature note. Advection–attenuation (first-order transport, Beer–Lambert type) and characteristic methods: Evans [10].

4.3.3 Entropic–gravity corrections (attenuation, production, capacity)

Setup. A radial boundary in n spatial dimensions has area $A_n(r) = S_{n-1} r^{\,n-1}$. The radial flux density is $j_r(r)$; the directed flux is $\Phi(r) := A_n(r)\, j_r(r)$; the informational field amplitude is $E(r) := |\mathbf{E}_I(r)|$. Inward/outward components use the directional split. Window and frame notions are as defined in the axioms. The boundary capacity is defined by $N(r)$ addressable trits, a per-trit reliable rate $f_{\max}$, and the relation $N(r) = \alpha\, A_n(r)$.

Derivation. From the local continuity identity (2.5) and the Linear Channel Closure (2.9) ($j_r = -\kappa E$), one obtains the spatial balance and its radial reductions below.

Window balance (cut form). On a window $[t_0, t_1]$ over the worldtube between two nested spheres $S(r_1) \subset S(r_2)$, the cut identity (2.4) yields

$$\Delta I_{\mathrm{enc}} \;=\; N_{\mathrm{in}}(r_1) \;-\; N_{\mathrm{out}}(r_2) \;+\; \Pi_{[r_1,r_2]}, \qquad N_{\mathrm{in/out}}(r) := \int_{t_0}^{t_1}\!\!\int_{S(r)} \big(\mathbf{j}\cdot\mathbf{n}\big)_{\mp}\, \mathrm{d}A\, \mathrm{d}t. \tag{4.33}$$

For advective transport, (4.30) relates inlet and outlet histories by

$$\Phi_{\mathrm{out}}(t) \;=\; S\,\Phi_{\mathrm{in}}(t-\tau), \qquad S = \exp\!\Big(-\int \mu\, \mathrm{d}s\Big). \tag{4.34}$$

Frame reduction (steady limit). On a frame with $\partial_t i = 0$ and radially symmetric data, the local form (2.5) gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\,\Phi(r) \;=\; A_n(r)\,\sigma(r). \tag{4.35}$$

With $j_r = -\kappa E$,

$$\frac{\mathrm{d}}{\mathrm{d}r}\Big(A_n(r)\,E(r)\Big) \;=\; -\frac{1}{\kappa}\,A_n(r)\,\sigma(r). \tag{4.36}$$

Attenuation (sink). For a local removal rate $\mu(r) \geq 0$ with sink $\sigma_{\mathrm{sink}}(r) = -\mu(r)\,j_r(r)$, (4.36) and $j_r = -\kappa E$ yield

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}r}\Big(A_n E\Big) + \mu(r)\,A_n\,E = 0} \tag{4.37}$$

with solution

$$E(r) = \frac{C}{A_n(r)}\,\exp\!\Big(-\int_{r_0}^{r}\mu(u)\,\mathrm{d}u\Big), \qquad C = \frac{\Phi(r_0)}{\kappa}. \tag{4.38}$$

Volumetric source/sink (forcing). With an additional bounded source density $\sigma_{\mathrm{src}}(r)$ so that $\sigma = \sigma_{\mathrm{src}} - \mu\, j_r$,

$$\frac{\mathrm{d}}{\mathrm{d}r}\big(A_n E\big) + \mu\,A_n\,E \;=\; -\frac{1}{\kappa}\,A_n\,\sigma_{\mathrm{src}}, \tag{4.39}$$

and the integrating-factor solution is

$$E(r) = \frac{1}{A_n(r)}\,e^{-\int_{r_0}^{r}\mu}\left[C - \frac{1}{\kappa}\int_{r_0}^{r} A_n(\rho)\,\sigma_{\mathrm{src}}(\rho)\,e^{\int_{r_0}^{\rho}\mu}\,\mathrm{d}\rho\right]. \tag{4.40}$$

3D specialization. For $n = 3$, $A_3(r) = 4\pi r^2$,

$$E(r) = \frac{C}{4\pi r^2}\,\exp\!\Big(-\int_{r_0}^{r}\mu(u)\,\mathrm{d}u\Big). \tag{4.41}$$

Constant and power-law cases:

$$\mu(r) = 0:\; E = \frac{C}{4\pi r^2}; \qquad \mu(r) = \mu_0:\; E = \frac{C}{4\pi r^2}e^{-\mu_0(r-r_0)}; \qquad \mu(r) = \frac{a}{r}:\; E = \frac{C}{4\pi}\,\frac{r_0^a}{r^{2+a}}.$$

Capacity ceiling (inner flattening). From the setup definitions, $J_A(r) \leq f_{\max}N(r)$ and $j_r(r) \leq f_{\max}N(r)/A_n(r)$. With $N(r) = \alpha\,A_n(r)$,

$$E(r) \;\leq\; \frac{f_{\max}\alpha}{\kappa} \;=:\; E_{\max}. \tag{4.42}$$

In $n = 3$ the unattenuated profile $E = C/(4\pi r^2)$ intersects the ceiling at

$$r_{\mathrm{flat}} \;=\; \sqrt{\frac{C}{4\pi\,E_{\max}}}, \qquad E(r) \;=\; \min\left\{\,\frac{C}{4\pi r^2}\,e^{-\int_{r_0}^{r}\mu},\;\; E_{\max}\,\right\}. \tag{4.43}$$

Piecewise media and composition. For segments $\{(\mu_k, \sigma_{\mathrm{src},k}, r_k \to r_{k+1})\}$ with piecewise-constant data, survival factors multiply as $e^{-\mu_k \Delta r_k}$ and forced contributions add via (4.40). The ceiling (4.42) is applied per segment when $N(r)$ or $f_{\max}$ changes.

Potential drop (frame). From $\mathbf{E}_I = -\nabla \Phi_I$ in (2.9),

$$\Phi_I(r_2) - \Phi_I(r_1) \;=\; -\int_{r_1}^{r_2} E(\rho)\, \mathrm{d}\rho. \tag{4.44}$$

Sectors and confinement. For a boundary sector of solid angle $\Delta\Omega$, the sector area is $A_\Delta(r) = \Delta\Omega\, r^{\,n-1}$; relations (4.37)–(4.44) hold with $A_n \mapsto A_\Delta$. In effective 1D (guides) or 2D (sheets), the same relations hold with the constant cross-section or circumference.

Falsification. (i) In a source-free annulus with $\mu = 0$, measured $E(r)$ must follow $r^{-(n-1)}$; persistent departures falsify (4.41) (specialized). (ii) With known $\mu(r)$, the ratio $E(r)A_n(r)\, e^{\int \mu}$ must be constant (from (4.37)); deviations falsify the attenuation premise. (iii) In a sector with fixed $\alpha, f_{\max}$, failure to observe a plateau at r_{flat} computed from (4.43) falsifies the ceiling assumption on that window.

Provenance. (2.5), (2.9), (4.30), (4.35)–(4.43).

Literature note. Linear transport/attenuation and integrating factors: Evans [10]. Capacity and reliable communication limits: Shannon [4]; Cover & Thomas [11].

4.4 Mechanics

4.4.1 Thermodynamics to Newtonian rate laws (constant-free)

Setup. On configuration–time space (q, t) ⋆2.2.3, assume a quadratic kinetic term with a positive-definite, window-constant mass metric M and an external potential:

$$H(q, P, t) = \tfrac{1}{2}\, P \cdot M^{-1} P \;+\; U(q, t), \qquad M > 0. \tag{4.45}$$

Derivation. From the canonical rate equations (2.8) applied to (4.45), $\dot{q} = M^{-1}P$, $\dot{P} = -\partial_q U$. With $\mathbf{p} := M\dot{q}$ and $\mathbf{f} := -\,\partial_q U$, one obtains the Newtonian rate law

$$\dot{\mathbf{p}} \;=\; \mathbf{f}. \tag{4.46}$$

Specialization. Work–power identity for $K(P) := \frac{1}{2}\, P \cdot M^{-1} P = \frac{1}{2}\, \dot{q} \cdot M\dot{q}$:

$$\frac{dK}{dt} = \dot{q} \cdot \dot{P} = \dot{q} \cdot \mathbf{f} = \mathbf{f} \cdot \mathbf{v}, \qquad \mathbf{v} := \dot{q}. \tag{4.47}$$

If U is time-independent, $\dot{H} = 0$ for $H = K + U$.

Calibration. With the one-datum linear sector mapping (A.4) and scale (A.5), measured sector fields satisfy

$$\mathbf{g} = c_M\, \mathbf{E}_I, \qquad \mathbf{E} = c_Q\, \mathbf{E}_I. \tag{4.48}$$

Falsification. In a constant-mass, time-independent U regime, if measured $\mathbf{p}(t)$ and $\mathbf{f}(t)$ on the same window fail $\dot{\mathbf{p}} = \mathbf{f}$ by more than the declared estimation bounds, either the Hamiltonian choice (4.45) or the calibration (4.48) is inconsistent with the axioms invoked.

Provenance. (2.8), (4.45), (4.46), (4.47), (4.48), (A.4), (A.5).

Literature note. Poincaré–Cartan form and Hamiltonian rates: Arnold [12]. Convex duality and Legendre regularity: Rockafellar [13].

4.4.2 Least-action equivalence

Setup. On (q,t) as in ⋆2.2.3, the canonical rate equations (2.8) hold. Legendre regularity (strict convexity in P) is assumed on the sector.

Derivation. Define the Legendre pair

$$\mathcal{R}(q,\dot q,t) \;:=\; \sup_{P}\,\big(P_i\dot q^i - H(q,P,t)\big), \qquad H(q,P,t)=\sup_{\dot q}\,\big(P_i\dot q^i-\mathcal{R}(q,\dot q,t)\big). \tag{4.49}$$

Using the Poincaré–Cartan form $\Theta_{\mathcal{R}} = P_i\,dq^i - H\,dt$ ⋆2.2.3, closure on rectangular Σ implies $\int_{\partial\Sigma}\Theta_{\mathcal{R}}=0$. Hence the variation of

$$\mathcal{S}[q] \;:=\; \int_{t_0}^{t_1}\mathcal{R}\big(q,\dot q,t\big)\,dt \tag{4.50}$$

vanishes for fixed endpoints, $\delta\mathcal{S}[q]=0$. Therefore the Euler–Lagrange equations

$$\frac{d}{dt}\Big(\frac{\partial\mathcal{R}}{\partial\dot q^i}\Big) \;-\; \frac{\partial\mathcal{R}}{\partial q^i} \;=\; 0 \tag{4.51}$$

are equivalent to the canonical rates (2.8).

Specialization. At the stationary pair, $\dot q^i=\partial_{P_i}H$ and $P_i=\partial_{\dot q^i}\mathcal{R}$; with $\mathcal{R}=P\cdot\dot q-H$, one recovers $\dot P_i=-\partial_{q^i}H$.

Falsification. On a short window with fixed endpoints, if $\int_{\partial\Sigma}\Theta_{\mathcal{R}}\neq 0$ under the declared regularity and boundary conditions, the equivalence between (4.51) and (2.8) is violated for the modelling choice at hand.

Provenance. (2.7), (2.8), (4.49)–(4.51).

Literature note. Equivalence of Hamilton/Euler–Lagrange via the Poincaré–Cartan form: Arnold [12]; geometric mechanics (Abraham–Marsden) [14]; convex duality and Legendre transforms: Rockafellar [13].

4.5 Electrodynamics

4.5.1 Info–EM (Maxwell form)

Setup. On a wave window with the wave–supporting closure ((2.10)–(2.12)), choose an informational potential A_I (a 1–form) and define the field 2–form $F_I := dA_I$. Let the excitation be $H_I := \chi : F_I$ (medium/model choice). Boundary orientation is as in §2.1.

Derivation. From $F_I = dA_I$ (identity of exterior calculus), $dF_I=0$. With $H_I=\chi:F_I$, define the current 3–form $J_I := dH_I$; then $dH_I=J_I$ and $dJ_I=0$, consistent with the continuity/Stokes family ((2.1)–(2.5)). The Maxwell system on the window is

$$\boxed{dF_I=0, \qquad dH_I=J_I} \tag{4.52}$$

with $H_I=\chi:F_I$ supplying the constitutive relation.

Specialization. Under a (3+1) split identifying $F_I \leftrightarrow (\mathbf{E}, \mathbf{B})$ and $H_I \leftrightarrow (\mathbf{D}, \mathbf{H})$, (4.52) reduces to

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \quad \nabla \cdot \mathbf{D} = \rho, \quad \nabla \times \mathbf{H} - \partial_t \mathbf{D} = \mathbf{J}, \tag{4.53}$$

where $J_I \leftrightarrow (\rho, \mathbf{J})$. Poynting's theorem is the instance of ((2.11)–(2.12)) with $u = \frac{1}{2}(\mathbf{E}\cdot\mathbf{D} + \mathbf{B}\cdot\mathbf{H})$ and $\mathbf{S} = \mathbf{E} \times \mathbf{H}$.

Literature note. Forms & Stokes: Frankel; macroscopic Maxwell: Jackson; Landau–Lifshitz–Pitaevskii.

4.6 Geometry & Dimensional Tools

4.6.1 Geometry as tool—flat ↔ curved

Setup. On a (Lorentzian or Riemannian) manifold, a future-directed null or timelike congruence k^a threads a narrow tube of trajectories, with affine (or proper) parameter λ and cross-sectional area $A(\lambda)$ orthogonal to k^a.

Expansion and area. The expansion is

$$\theta \;:=\; \nabla_a k^a, \qquad \frac{d}{d\lambda} \ln A \;=\; \theta. \tag{4.54}$$

From the tube invariance (4.2) on steady, source-free segments, Φ is constant and $\langle j_n \rangle = \Phi/A$, hence

$$\frac{d}{d\lambda} \ln\langle j_n \rangle \;=\; -\,\theta. \tag{4.55}$$

With the wave envelope in homogeneous segments (3.2), $\langle |\mathbf{E}_I| \rangle \propto A^{-1/2}$, giving

$$\frac{d}{d\lambda} \ln\langle |\mathbf{E}_I| \rangle \;=\; -\,\frac{1}{2}\,\theta. \tag{4.56}$$

Raychaudhuri focusing. For a geodesic, hypersurface-orthogonal congruence ($\omega_{ab} = 0$), the Raychaudhuri equation is

$$\frac{d\theta}{d\lambda} \;=\; -\,\frac{1}{2}\,\theta^2 \;-\; \sigma_{ab}\sigma^{ab} \;-\; R_{ab}\,k^a k^b, \tag{4.57}$$

where σ_{ab} is the shear and R_{ab} the Ricci tensor. For timelike congruences, the coefficient of θ^2 is dimension-dependent; the sign of the Ricci term is unchanged.

Calibrated equality. On the same region used for sector calibration, the focusing term relates to a calibrated stress component by

$$R_{ab}\,k^a k^b \;\overset{\text{cal}}{=}\; 8\pi\,\mathcal{T}^{(\text{cal})}_{ab}\,k^a k^b. \tag{4.58}$$

Flat specialization. In flat geometry with a radial congruence in n spatial dimensions, $A(\lambda) \propto r^{\,n-1}$ and $\theta = (n-1)\,\dot{r}/r$. Then (4.55) yields $\langle j_n \rangle \propto r^{-(n-1)}$, and (4.56) yields $\langle |\mathbf{E}_I| \rangle \propto r^{-(n-1)/2}$.

Shear and vorticity. Shear σ_{ab} modifies $A(\lambda)$ through (4.57); nonzero vorticity ω_{ab} counteracts focusing. The amplitude rules (4.55)–(4.56) follow from the resulting $A(\lambda)$.

Literature note. Geodesic congruences and Raychaudhuri: Wald [15], Raychaudhuri [16], Poisson [17]. Local-horizon focusing: Jacobson [18].

Table 1: Geometric scaling of frame amplitudes and wave envelopes by effective spatial dimension n_{eff}.

Effective dimension	**Area** $A_n(r)$	**Frame** $\lvert\mathbf{E}_I\rvert$	**Wave** $\langle\lvert\mathbf{E}_I\rvert\rangle$
1D (guide / tube)	$A_\perp$ (const.)	$\propto r^0$	$\propto r^0$
2D (sheet / interface)	$A_2(r) \propto r$	$\propto r^{-1}$	$\propto r^{-1/2}$
3D (spherical)	$A_3(r) = 4\pi r^2$	$\propto r^{-2}$	$\propto r^{-1}$

4.6.2 Dimensional flows & confinement

Setup. A *frame* App. B is taken with $\partial_t i = 0$ on an open annulus $W \subset U$ where $\sigma|_W = 0$. In n spatial dimensions, a radius–r boundary has area $A_n(r)$; a conical sector of solid angle $\Delta\Omega$ has $A_\Delta(r) = \Delta\Omega\, r^{\,n-1}$.

Frame scaling. From tube invariance (4.2) and the Linear Channel Closure (2.9),

$$\big\langle j_n \big\rangle(r) = \frac{\Phi}{A_n(r)}, \qquad \big|\mathbf{E}_I(r)\big| = \frac{1}{\kappa}\,\frac{\Phi}{A_n(r)}. \tag{4.59}$$

For spherical or conical boundaries with $A_n(r) = S_{n-1}\, r^{\,n-1}$, this gives $|\mathbf{E}_I| \propto r^{-(n-1)}$.

Wave envelope. On source-free windows in homogeneous segments, exported power is radius-independent and the far-field envelope (3.2) satisfies

$$\big\langle |\mathbf{E}_I(r)| \big\rangle \;\propto\; \big[A_n(r)\big]^{-1/2}; \tag{4.60}$$

for spherical/conical boundaries this yields $\langle|\mathbf{E}_I|\rangle \propto r^{-(n-1)/2}$.

Dimensional summary (table).

Confinement (effective dimension). Boundaries that constrain propagation reduce the accessible dimension n_{eff}.

- **Waveguides (effective 1D).** Cross-section $A_\perp$ fixed along the axis: $\langle j_n\rangle = \Phi/A_\perp$ and $|\mathbf{E}_I| = \Phi/(\kappa A_\perp)$ are independent of r; wave envelopes remain r^0.
- **Sheets / interfaces (effective 2D).** Fixed thickness implies $A_2(r) \propto r$: frame amplitudes scale as r^{-1}; wave envelopes scale as $r^{-1/2}$.

Directional sectors. For an aperture subtending $\Delta\Omega$ in n dimensions, the sector area is $A_\Delta(r) = \Delta\Omega\, r^{\,n-1}$. Relations (4.59)–(4.60) hold with $A_n \mapsto A_\Delta$.

Literature note. Geometric flows and inverse-power laws: Frankel [1]; radiation and far-field envelopes: Jackson [2]; Landau, Lifshitz & Pitaevskii [3].

4.7 Extensions

4.7.1 Quantum path–integral template (from informational fluctuations)

Setup. On configuration–time space (q, t), the canonical rate equations and Poincaré–Cartan form hold by (2.8) and (2.7). On a window $[t_0, t_1]$, fluctuations about the canonical drift are modeled by local quadratic data: a drift $v(q,t)$, diffusivity $D(q,t) > 0$, scalar rate $U(q,t)$, and an inertial map $M(q,t) > 0$. A positive wave scale $\alpha > 0$ is used in the oscillatory branch.

Derivation. From (2.8) (drift) and a quadratic deviation penalty, the diffusive branch uses the Onsager–Machlup density

$$\mathcal{R}_{\text{diff}}(q,\dot{q},t) \;=\; \frac{1}{2}\,(\dot{q}-v)^{T}D^{-1}(\dot{q}-v) \;+\; U(q,t). \tag{4.61}$$

The associated path weight is

$$\mathbb{P}[\gamma] \;\propto\; \exp\!\left(-\int_{t_0}^{t_1}\mathcal{R}_{\text{diff}}\,dt\right), \tag{4.62}$$

and the state density satisfies the Fokker–Planck equation

$$\partial_t\rho \;=\; -\nabla\cdot(v\,\rho) \;+\; \tfrac{1}{2}\,\nabla^{T}\big(D\,\nabla\rho\big). \tag{4.63}$$

For an oscillatory branch based on the rate 1-form, the path amplitude uses

$$\mathcal{A}[\gamma] \;\propto\; \exp\!\left(i\,2\pi\int_{t_0}^{t_1}\big[P\cdot\dot{q}-H(q,P,t)\big]\,dt\right), \tag{4.64}$$

with quadratic generator

$$H(q,P,t) = \tfrac{1}{2}\,P^{T}M^{-1}(q,t)\,P \;+\; U(q,t). \tag{4.65}$$

A short-time kernel (Fourier representation) yields the Schrödinger-type evolution

$$i\,\partial_t\psi \;=\; \Big[-\;\tfrac{\alpha^2}{2}\,\nabla^{T}\big(M^{-1}\nabla\big)+U(q,t)\Big]\psi. \tag{4.66}$$

Minimal coupling to an informational 1-form A_I is

$$P \;\mapsto\; P-A_I(q,t) \quad \text{in} \quad (4.64)(4.66). \tag{4.67}$$

Specialization. For constant M and D, (4.63) reduces to drift–diffusion with uniform coefficients, and (4.66) becomes a constant-mass Schrödinger generator.

Calibration. (omitted)

Falsification. On a declared window with fixed (v, D, U, M), if the measured transition densities deviate from (4.63) beyond statistical tolerance on the same window, the diffusive branch is falsified. If interferometric phase transport fails to obey (4.66) under the same U, M, α, the oscillatory branch is falsified.

Provenance. (2.7), (2.8), (4.61)–(4.67).

Literature note. Onsager–Machlup; Fokker–Planck (Risken; Gardiner). Path integrals (Feynman; Feynman–Hibbs).

4.7.2 Confinement potentials

Setup. Frame form with $\partial_t i = 0$ on an open annulus $W \subset U$ where $\sigma|_W = 0$. Let $A_n(r)$ be the area of a radius–r boundary in n spatial dimensions. Linear channel closure holds by (2.9).

Derivation. From the frame-boundary relation (4.1.2) and (2.9),

$$\big|E_I(r)\big| \;=\; \frac{1}{\kappa}\,\frac{\Phi}{A_n(r)}. \tag{4.68}$$

The potential drop along a radial segment satisfies

$$\Phi_I(r_2)-\Phi_I(r_1) \;=\; -\int_{r_1}^{r_2}E(\rho)\,d\rho, \tag{4.69}$$

by (4.44).

Specialization. *(1D tube)* If $A_1(r) = A_\perp$ is constant,

$$E(r) = E_{\text{tube}} := \frac{\Phi}{\kappa\, A_\perp}. \tag{4.70}$$

(3D spherical, unsaturated) If $A_3(r) = 4\pi r^2$,

$$E(r) = \frac{\Phi}{4\pi\kappa}\,\frac{1}{r^2}. \tag{4.71}$$

(Ceiling and flattening) With the capacity ceiling (4.42), the intersection radius in $n = 3$ is

$$r_{\text{flat}} = \sqrt{\frac{\Phi/\kappa}{4\pi\, E_{\max}}}, \quad E(r) = \min\left\{\frac{\Phi}{4\pi\kappa r^2},\ E_{\max}\right\}. \tag{4.72}$$

Calibration. With a one-datum sector calibration $U = c_{\text{conf}}\Phi_I$, *(1D)* the separation potential is

$$U(r) = U_0 + \sigma r, \qquad \sigma = c_{\text{conf}}\, E_{\text{tube}}. \tag{4.73}$$

(3D + ceiling) using (4.72) and (4.69),

$$U(r) = U_0 + c_{\text{conf}} \begin{cases} E_{\max}\, r, & r \le r_{\text{flat}}, \\ E_{\max}\, r_{\text{flat}} + \dfrac{\Phi}{4\pi\kappa}\left(\dfrac{1}{r_{\text{flat}}} - \dfrac{1}{r}\right), & r > r_{\text{flat}}. \end{cases} \tag{4.74}$$

Falsification. (1D) In a declared guide, if $U(r)$ departs from linearity by $> 10^{-4}$ over 10^3 wavelengths (on the same segment), the 1D specialization is falsified. (3D) If the calibrated $E(r)$ fails the $1/r^2$ law outside the declared production region, either $\sigma \neq 0$ on W or the ceiling assumption is violated.

Provenance. (4.1.2), (2.9), (4.44), (4.42).

Literature note. Inverse-power radial fields and sector areas: Jackson. Geometric flux/Stokes: Frankel.

4.7.3 QCD string (declared 1D tube)

Setup. Frame form with $\partial_t i = 0$ on an open $W \subset U$ where $\sigma|_W = 0$. Declare an effective 1D flux-tube with constant cross-section $A_\perp$. Adopt the linear channel closure (2.9).

Derivation. From (4.70), the informational field inside the tube is uniform:

$$E_I \;=\; \frac{\Phi}{\kappa\, A_\perp} \;=:\; E_{\text{string}}. \tag{4.75}$$

With $\mathbf{E}_I = -\nabla\Phi_I$ and a sector calibration $U = c_{\text{conf}}\Phi_I$, the separation potential is

$$U(r) = U_0 + \sigma r, \qquad \sigma = c_{\text{conf}}\, E_{\text{string}}. \tag{4.76}$$

Specialization. For a lattice-anchored $A_\perp = \pi r_0^2$, any single datum for σ fixes $c_{\text{conf}}E_{\text{string}}$.

Calibration. A typical datum $\sigma \approx (0.44\,\text{GeV})^2$ sets the slope in the chosen sector.

Falsification. If the measured force $F = dU/dr$ differs from a constant by $> 3\%$ for $r \gtrsim 1.5\,\text{fm}$ in the declared 1D regime, the tube specialization is falsified.

Provenance. (2.9), (4.70), (4.75)–(4.76).

Literature note. External provenance only: lattice flux-tube/string-tension measurements (Wilson loops; Polyakov-type formulations).

Appendices

This appendix set collects supporting material used in the main text:

1. App. A Kick (Kz) units & Kram anchors
2. App. B Glossary & Symbols

A Kick (Kz) units & Kram anchors

Scope. Records the Kz unit convention (division by h), an optional Kram anchor ladder, one-datum sector calibrations, and the capacity/erasure relations used in the body. This appendix is reference-only; it introduces no new axioms and does not restate cap/side formulas.

Setup. Kz units. For any SI quantity X, define $X_{\mathrm{Kz}} := X/h$ using Planck's constant h (not $\hbar$).
Count base. Information counts default to trits with $S_{\mathrm{nats}} = (\ln 3)\, S_{\mathrm{trits}}$ and $S_{\mathrm{bits}} = S_{\mathrm{nats}}/\ln 2$.
Screens/windows. $A_n(r)$ denotes a geometric $(n-1)$-screen at scale r. A measurement window has duration Δt.
Local notation. $\Phi(r) := \int_{A_n(r)} (\mathbf{j}\cdot\mathbf{n})\,\mathrm{d}A$ (tube flux through $A_n(r)$); $\overline{J}_A := \frac{1}{A}\int_A |\mathbf{j}\cdot\mathbf{n}|\,\mathrm{d}A$ (admissible area-averaged boundary entropic-flux bound over patch A); κ (proportionality constant from the frame boundary relation); $\mathbf{E}_I$ (instrument reference field used for linear readouts).

Statement(s).

$$E_{\mathrm{z}} = \frac{E}{h}, \qquad m_{\mathrm{z}} = \frac{mc^2}{h}, \qquad T_{\mathrm{z}} = \frac{k_B T}{h}, \qquad P_{\mathrm{z}} = \frac{P}{h}, \tag{A.1}$$

where E_{z}, m_{z} (rest-energy rate), and T_{z} have units s^{-1}, and P_{z} has units s^{-2}. Hence $P_{\mathrm{z}}/T_{\mathrm{z}}$ has units s^{-1}.

$$\frac{\Delta I_{\mathrm{trits}}}{\Delta t} \;\le\; \frac{P_{\mathrm{z}}}{T_{\mathrm{z}}\,\ln 3}, \tag{A.2}$$

with equivalent forms $\Delta I_{\mathrm{nats}}/\Delta t \le P_{\mathrm{z}}/T_{\mathrm{z}}$ and $\Delta I_{\mathrm{bits}}/\Delta t \le P_{\mathrm{z}}/(T_{\mathrm{z}}\ln 2)$.

$$1\,k = 10^{47}\,\mathrm{Kz}, \qquad 1\,kk = 10^{50}\,\mathrm{Kz}, \qquad 1\,Mk = 10^{53}\,\mathrm{Kz}, \qquad 1\,Gk = 10^{56}\,\mathrm{Kz}, \qquad 1\,Tk = 10^{59}\,\mathrm{Kz}. \tag{A.3}$$

Note. These anchors are convenience scales; they do not modify the Kz units $X_{\mathrm{Kz}} = X/h$.

$$\mathbf{g} = c_M\,\mathbf{E}_I, \qquad \mathbf{E} = c_Q\,\mathbf{E}_I, \qquad \mathbf{j}_{\mathrm{th}} = \kappa_T\,\mathbf{E}_I, \tag{A.4}$$

where the sector constant $c_{\mathrm{sec}} \in \{c_M, c_Q, \kappa_T\}$ is fixed by one datum in that sector.

$$c_{\mathrm{sec}} = \frac{Y_\star}{|\mathbf{E}_I(r_\star)|} = Y_\star\,\frac{\kappa\,A_n(r_\star)}{\Phi_\star}, \qquad Y_\star \in \{g_\star, E_\star, j_{\mathrm{th},\star}\}, \quad \Phi_\star := \Phi(r_\star), \tag{A.5}$$

with $A_n(r)$ the screen area at scale r and $\Phi(r)$ the tube flux through it.

$$Y(r) = c_{\mathrm{sec}}\,|\mathbf{E}_I(r)| = Y_\star\,\frac{\Phi(r)}{\Phi_\star}\,\frac{A_n(r_\star)}{A_n(r)}. \tag{A.6}$$

$$\frac{\Delta I_{\mathrm{trits}}}{\Delta t} \;\le\; \min\!\left\{\frac{A\,\overline{J}_A}{\ln 3},\quad C^{(\mathrm{trits})}_{\mathrm{chan}}(\varepsilon),\quad \frac{P_{\mathrm{z}}}{T_{\mathrm{z}}\,\ln 3}\right\}, \tag{A.7}$$

where $C^{(\mathrm{trits})}_{\mathrm{chan}}(\varepsilon) = \mathcal{C}\big[R^{(a)};\,\mathcal{N},\,\mathcal{B};\,\varepsilon\big]$ denotes a reliable channel capacity for boundary controls $R^{(a)}$, noise kernel $\mathcal{N}$, and budgets $\mathcal{B}$ (including P_{z}, T_{z}, area A, window duration Δt), at reliability ε.

Specializations (optional).

$$\text{Lossless frame: } \Phi(r) = \Phi_\star \;\Rightarrow\; Y(r) = Y_\star\,\frac{A_n(r_\star)}{A_n(r)} \;\propto\; A_n(r)^{-1}. \tag{A.8}$$

Notation / units (if needed). Exact constants for quick checks: $h = 6.62607015 \times 10^{-34}\,\mathrm{J\,s}$, $\dfrac{k_B}{h} = 2.083661912 \times 10^{10}\,\mathrm{s^{-1}\,K^{-1}}$, $\dfrac{1\,\mathrm{eV}}{h} = 2.417989242 \times 10^{14}\,\mathrm{s^{-1}}$, $\dfrac{1\,\mathrm{W}}{h} = 1.509190179 \times 10^{33}\,\mathrm{s^{-2}}$. Typical conversions: $E[\mathrm{J}] \mapsto E/h$ $(\mathrm{s^{-1}})$, $E[\mathrm{eV}] \mapsto (E\,e)/h$ $(\mathrm{s^{-1}})$, $T[\mathrm{K}] \mapsto (k_B T)/h$ $(\mathrm{s^{-1}})$, $P[\mathrm{W}] \mapsto P/h$ $(\mathrm{s^{-2}})$.

Literature note. Shannon (1948); Cover & Thomas (2006) for information and reliable communication. Landauer (1961); Bennett (2003) for erasure cost and reversibility.

B Glossary & Symbols

Glossary of agnostic terms

Window (definition). A *window* is a finite interval $[t_0, t_1]$ on a fixed region U. Window statements apply on $U \times [t_0, t_1]$ using the integral and cut forms (2.3)–(2.4).

Frame (definition). A *frame* is a time-frozen spatial slice $V_t := U \cap \{t = \text{const}\}$. The frame form of a window statement is the limit $\Delta t \to 0$ evaluated on V_t (cf. (2.5)).

Worldtube (cap/side; moving boundary). The time-extended boundary $\partial(U \times [t_0, t_1])$ decomposes into a top cap, a bottom cap, and a side. For moving boundaries with normal speed v_n, the side integrand used in balances is

$$\mathbf{j}\cdot\mathbf{n} \;-\; i\, v_n.$$

Directional split. For any scalar a, define $a_\pm := \frac{1}{2}(a \pm |a|)$. Inward/outward boundary fluxes use $(\mathbf{j}\cdot\mathbf{n})_-$ and $(\mathbf{j}\cdot\mathbf{n})_+$.

Boundary identification and heat mapping (pointer). On boundaries, the informational current j is identified with the entropy current j_S, the density i with s, and the heat flux is $q := T_{\mathrm{z}}\, j_S$. The mapping is the axiom (2.2) in ⋆2.

Heat on a window (total). On a boundary patch $A \subset \partial U$ over $[t_0, t_1]$, the total heat is defined and used in the body as $Q_A([t_0, t_1])$ in (4.20).

Heat on a frame (flux density). In frame form, the heat-flux density is $\mathbf{q} = T_{\mathrm{z}}\, \mathbf{j}_S$ (cf. (4.21)).

List of Symbols (reference)

j	Informational/entropic current (1-form). Identity: (2.1).
j_S	Entropy current on boundaries; $j \equiv j_S$ by (2.2).
q	Heat-flux density $q = T_{\mathrm{z}}\, j_S$. Frame form in (4.21).
i	Informational/entropic density (nats/vol). Local law: (2.5).
$\Sigma,\ \sigma$	Production n-form $\Sigma = \sigma\,\mathrm{vol}$ and density σ. Identity: (2.1).
J^μ	Coordinate current; $\partial_\mu J^\mu = \sigma$. Identity: (2.1).
$\Phi_{\mathrm{in/out}}$	Directed boundary fluxes (directional split). Cut form: (2.4).
I_{enc}	Enclosed information. Cut form context: (2.4).
Φ	Tube/sector flux through a screen. Definition: (4.1).
$\mathbf{E}_I$	Informational field amplitude; closure: (2.9).
Φ_I	Informational potential with $\mathbf{E}_I = -\nabla\Phi_I$ in (2.9).
κ	Linear-channel coefficient; $\mathbf{j} = -\,\kappa\,\mathbf{E}_I$ in (2.9).
T_{z}	Boundary temperature in Kz units; (A.1).
P_{z}	Power budget in Kz units; (A.1).

P	Canonical momentum in the rate (Poincaré–Cartan) form (2.7).
μ	Attenuation/removal rate along a tube; transport: (4.28).
$A_n(r)$	$(n-1)$-dimensional boundary area; sector $A_\Delta(r) = \Delta\Omega\, r^{\,n-1}$.
$A_\perp$	Effective cross-section (guides/tubes).
θ	Expansion of a tube congruence.
J_A	Inward boundary count rate on patch A (window); (4.10).
$N,\ \alpha$	Addressable trits and areal density $N(r) = \alpha\, A_n(r)$.
$E_{\max}$	Ceiling field; (4.42).
Q_A	Total heat through patch A on a window; (4.20).

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